

TEACHING LEARNING IN MATHEMATICS

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4.1 INTRODUCTION

Children are inherently curious and explore the world by interacting with it. They love hands-on experiences with new things, regardless of their complexity. In fact, by engaging with them, they attempt to address complexity and discover their own solutions. This abundant curiosity leads to numerous questions and serves as the primary motivation for learning. Constructivists also advocate for children's self-construction of knowledge. They advocate creating an environment that fosters knowledge construction, especially when children are learning abstract mathematical concepts. But what does "knowledge construction" truly mean? Is it simply the teacher's explanation, taking notes, reading a book, or asking questions?

This lesson is specifically designed to enhance your pedagogical skills for teaching mathematics at the primary level. By engaging with the content

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herein, you will gain a comprehensive understanding of the innovative teaching approaches advocated by NEP 2020, including **Experiential Learning, the ELPS approach, and the four-block teaching method**. This will help you effectively **incorporate the ELPS and Four-Block Teaching approaches into your daily classroom practices** and confidently **utilize textbooks and other readily available materials** in your environment to make learning mathematics meaningful and engaging for your students.



4.2 LEARNING OUTCOMES

After completion of the chapter, the learner:

- identifies and explains the core approaches advocated by NEP 2020: Experiential Learning, ELPS, and the Four-Block Teaching method;
- incorporates the ELPS approach and Four-Block Teaching into their classroom practices; and
- utilises textbooks and other readily available materials for teaching mathematics meaningfully.

4.3 APPROACHES TO TEACHING MATHEMATICS AT THE FOUNDATION AND PREPARATORY LEVEL

4.3.1 Experiential Learning

Sudha, a Grade 2 teacher, wanted to teach three-digit addition to her class. Before introducing the topic, she gave students a few questions involving addition, both with and without regrouping (carrying over). While observing, she noticed that Radha solved all problems correctly using standard algorithms except one. She correctly added $67 + 24 = 91$, but when adding $48 + 7$, she answered 118.

- What do you think could be the reason for such a mistake?
- Is Sudha's approach appropriate as a teacher? Why?
- List three insights you, as a teacher, would gain from the above example.

Through this, Sudha aimed to assess students' prior knowledge and identify areas of misconception. Using these observations, she planned her next strategy. The next day, she had the class play "Snakes and Ladders" in small groups. On day three, she again used the "Snakes and Ladders" game, but this time, class discussion was an integral part of it. At times, she used bundles and sticks to address confusions.

- What could be the possible reasons for her planning this strategy? What issue did this resolve that further helped learners in their understanding?

Sudha was actually trying to provide her learners with a **hands-on experience**. She had opted for **experiential learning**, a process where learning occurs **by doing** rather than just receiving information. Including discussion allowed the children to reflect on their experience and perception, moving beyond mere activity to deeper understanding. The hands-on activity provided them with a **concrete experience**. Repeating the same activity with discussion allowed learners to **reflect** and articulate what they had dealt with, solidifying their learning.

Experiential learning, as proposed by Kolb, has four main pillars: **concrete experience, reflective observation, abstract conceptualization, and active experimentation**. (Morris, T.H (2020))

- **Concrete Experiences:** These refer to **hands-on opportunities** or “learning by doing.” The core principle here is that children construct knowledge on their own. Thus, we should provide opportunities where learners can build knowledge through **active participation** and explore all possible solutions. For example, while teaching shapes in Grade 4, giving children the opportunity to explore shapes and relate them to their daily life experiences is a wonderful example of concrete experience. Instead of just giving definitions and having them memorised, children can create something using shapes, which not only familiarizes them with shapes but also helps them build **spatial understanding**.
- **Reflection:** After exploring through experience, the next crucial aspect is to revisit the process by **reflecting on the procedure** one underwent. The main objective of this reflection is to mentally revisit the process and express it in one’s own words. This makes experiential learning **more comprehensive and precise**. It allows learners to analyze what worked and what didn’t. It also helps the scaffolder (teacher) to understand the learner’s thinking process. In the same chapter, after concrete experience, some questions are provided, and discussion is suggested for teachers. This discussion and these questions lead the child to reflect on the strategy they followed and to revisit how they made decisions.
- **Abstract Conceptualisation:** Learners develop new concepts and theories based on their reflections and observations, and then develop thinking in abstraction related to the experience. Here, the learner creates, formulates new ideas, and relates them to already existing concepts. After discussion, a project on making a model of a building is assigned to learners. What

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is the objective of assigning this project after concrete experience and discussion? The child gets the opportunity to reuse pre-existing knowledge in creating new ideas.

- **Active Experimentation:** This involves testing the new concepts and theories by applying them to new situations or evaluating the knowledge gained in different scenarios. Here, the learner applies their learning in various real-life situations while exploring different contexts. The various questions in the textbook provide such an opportunity.

Think and Reflect

Now that you've explored the principles of experiential learning, take a moment to reflect on your own teaching practice:

- How often do you currently incorporate hands-on activities that allow children to "do" mathematics?
- What opportunities do you provide for students to reflect on their mathematical experiences and articulate their understanding in their own words?



ACTIVITY 4.1

Consider a recent math lesson you taught. How could you redesign a part of it using the four pillars of experiential learning to make it more meaningful for your students?

4.3.2 ELPS

Aarti had to introduce division to Class 3 students. She reviewed the topic in the textbook to prepare herself. The next morning, in class, she began the chapter with the example discussed in the book:

“Hari has 15 pencils. He distributed them equally among his 5 friends. How many pencils does each of his friends get?”

As she posed the question, a few students answered immediately. Appreciating them, she asked, “How did you find the answer?” Very few children responded. Then, she declared that today she would teach how to divide, and using the chalkboard, she explained the algorithm of division.

According to you, is this the right way to introduce division at Grade 3? Why do you think so?

The most advocated approach these days, especially after the recommendations of NEP 2020, is the **ELPS approach**. The ELPS approach is built upon the firm belief that **a learner constructs knowledge most effectively when actively engaged**. This approach is designed to help children grasp mathematical concepts **gradually, deeply, and meaningfully** by moving from tangible experiences to abstract symbols (Lowrie T., 2018). It systematically builds understanding through four crucial stages:

- **E stands for Experience (Concrete):** This foundational stage refers to **hands-on exploration with concrete objects** to physically encounter and internalise a concept. This is where abstract ideas begin to take shape through direct interaction. For instance, to introduce place value, the prerequisite is grouping objects. Here, we provide opportunities for children to physically make groups or bundles of 10 using sticks, straws, an abacus, or other manipulatives. This ‘E’ of ELPS is about providing the **tangible, real-world connection** that forms the bedrock of understanding.
- **L stands for Language (Spoken/Verbal):** Language refers to **describing the experience in spoken form**. After actively making groups or bundles of 10, the teacher needs to initiate a discussion: “How did you make your bundles?” or “What did you observe when grouping these?” Answering such questions compels children to mentally revisit the process they followed, articulate their observations, and frame their concrete experiences into words. This stage is vital for developing mathematical vocabulary and for children to verbally connect their actions to the emerging concept. Further leading to questions like, “Can you use these bundles and sticks to represent numbers like 11, 20, etc.? How?”
 - What could be the expected answers from children? How is this helpful in learning the place value concept?
- **P Stands for Picture (Pictorial/Visual):** Here, the concept is **represented in a picture or visual form**. After experiencing and verbally articulating the concept, the next step is to translate it into drawings or visuals using charts, flashcards, or individual sketches in their notebooks. For example, “Show 23 using bundles and sticks.” Now, the child has to draw 2 bundles and 3 sticks. This process facilitates the **visual coding** of abstract ideas, like ‘2’ representing ‘two tens’ at the tens place, helping bridge the gap between concrete actions and abstract symbols.

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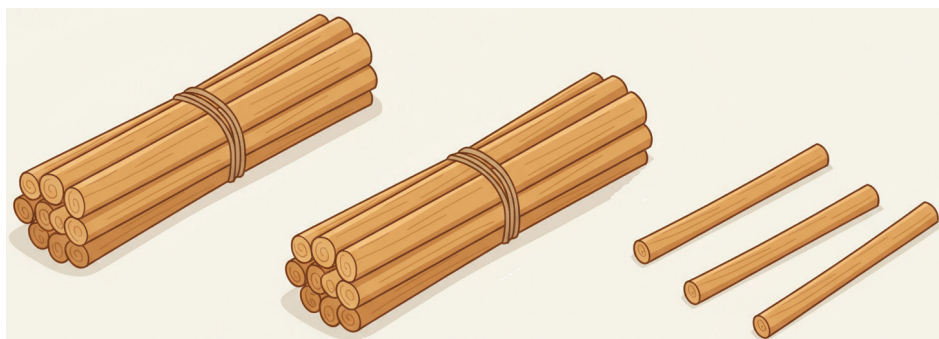
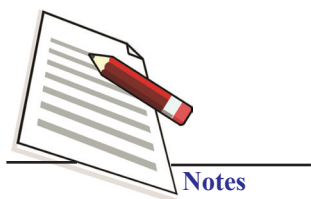


Fig 4.1 Representation of 23 using bundles and sticks

- S stands for Symbol (Abstract):** This is the final stage where the concept is introduced using **abstract symbols, numerals, or mathematical notations**. After having concrete experiences and pictorial representations, the learner is now ready to connect these rich understandings to the formal mathematical symbol. The symbol can be introduced alongside its pictorial representation, but it's crucial that it does not precede the concrete and pictorial stages. This systematic progression ensures that symbols are understood as concise representations of real-world experiences, not as isolated, meaningless characters.

ELPS is highly recommended for teaching mathematics because it follows a progression of modes of representation from concrete to picture and finally to symbol. Introducing concepts through tangible objects and visual representations significantly **reduces the inherent abstractness of mathematical concepts**, making them far more comprehensible and accessible for the learner.

In a typical classroom, abstract mathematical rules and concepts are often presented directly to learners. As mathematical concepts are abstract by nature, their direct presentation can make them difficult to grasp meaningfully, often promoting rote memorisation or superficial learning. Such pedagogies tend to revolve solely around the content, leaving little space for the learner's active construction of knowledge.

To develop abstract mathematical concepts meaningfully, they must be presented progressively: first in a concrete form, then pictorially, and finally in symbolic form, always integrating language throughout. This systematic exercise of ELPS not only reduces the abstraction of mathematical concepts but also provides a **sound and durable ground of understanding**. Many researchers advocate that most students are comfortable initially at the concrete stage or when concepts are presented using concrete objects. When coupled with

expressing their experiences in their own language, it not only builds vocabulary and language skills but also fosters metacognition. This progression offers the teacher a unique opportunity to understand the child's thinking process and how they arrived at their understanding. Here, learning systematically transfers from one mode to another, with language consistently playing a significant role in solidifying the connections.

Sidhi is teaching mathematics at the preparatory level. Let's see how Siddhi introduced division using ELPS to her Class 3 students.

Siddhi enters the class with a well-known activity, "*Bol Bhai Kitnay*" (Tell me, how many?). She took all the students to the ground and explained the game.

Siddhi said, "I'll ask '*Bol Bhai Kitnay*?' (How many?)."

Students: "*Aap bolo jitnay*." (You tell us the number.)

Siddhi: "I'll call out a number, say 8. All of you will make groups of 8. Those who are left out or make less than 8 in a group will be considered out from the game."

The next day, she carried buttons with her. She started the class with the same game, followed by a discussion about the procedure learners followed to solve the problem. After the discussion, she provided buttons to students. This time, she called out a number, and students had to distribute the buttons equally into that many groups. Then, they again engaged in a discussion.

The next day, the teacher changed her mode of presentation from concrete to picture. Using the blackboard, she drew a few pictures and asked students to distribute them into various groups. Students enjoyed the activity. Now, she again had a brief discussion on the "what" and "how" of the process.

The next day, along with pictorial representation, she introduced the symbols too.

Pictorial representation

(i) Divide among 3 children

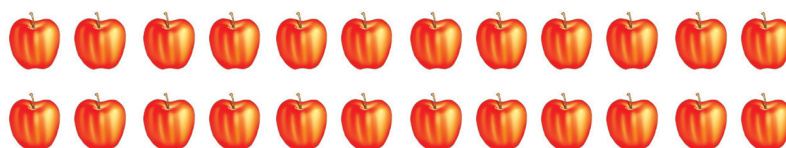


Fig 4.2

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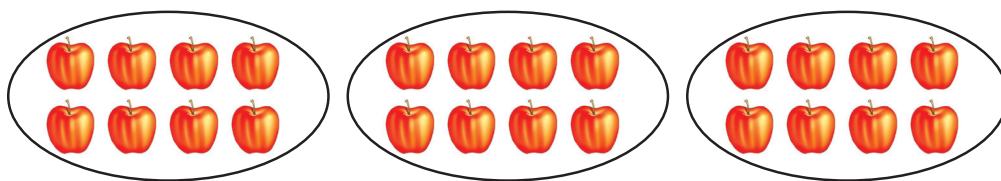


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3 groups of 8, so each child will get 8 apples

Fig 4.3

In symbol it is represented as $24 \div 3 = 8$.

And finally, on the fifth day, she introduced the standard procedure of division to the class.

Think & Reflect

What is your opinion on Siddhi's strategy?

Compare the pedagogy used by Aarti and Siddhi to teach division.



ACTIVITY 4.2

1. **Choose a specific mathematical concept** you teach at the primary level (e.g., multiplication, fractions, measurement, etc.).
2. Briefly **outline how you would introduce and develop this concept by moving through each stage of the ELPS approach:**
 - **E (Experience):** What concrete materials or hands-on activity would you use?
 - **L (Language):** How would you prompt students to describe their experience verbally? What key vocabulary would emerge?
 - **P (Picture):** How would you transition to representing the concept visually (e.g., drawings, diagrams, charts)?
 - **S (Symbol):** How and when would you introduce the formal mathematical symbol, linking it back to the concrete and pictorial representations?

4.3.3 Four-Block Approach

The **Four-Block Approach** is another method advocated in the NCF 2023 framework for teaching mathematics. It is designed to strengthen foundational numeracy skills. This approach encompasses four blocks of instruction: **oral**

math talk, skill teaching, skill practice, and math games. These four blocks comprise mathematical communication along with mathematical skills and thinking. This approach emphasizes a holistic learning approach in mathematics, focusing on conceptual understanding, procedural skills, and active engagement. Let's explore the four blocks.

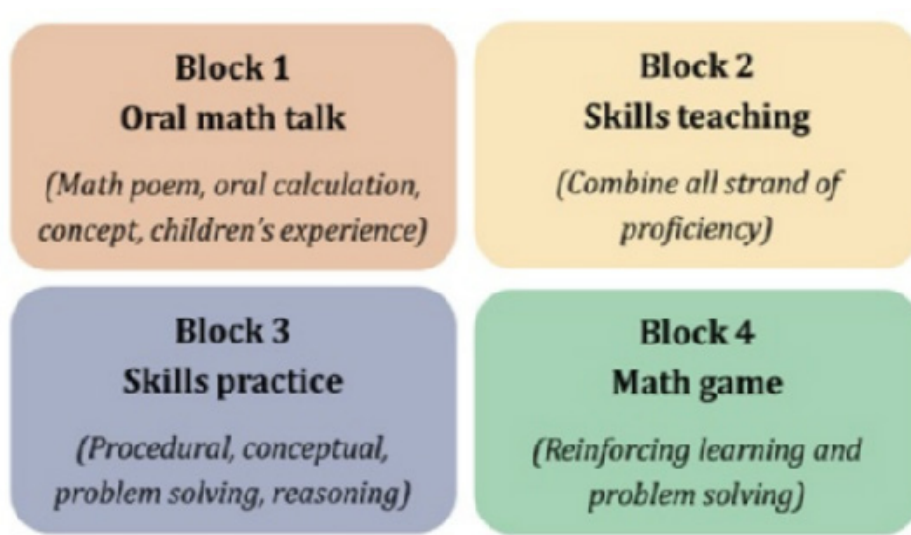


Fig 4.4 Components of four-block approach
 (National Curriculum Framework for Foundational Stage, Pg. 121)

Block 1: Oral Math Talk—Warming Up for Math!

Think of **Oral Math Talk** as the perfect warm-up for your math lesson. This block is all about engaging children and preparing their minds for new concepts. It's a fantastic way to link what they already know with what they're about to learn. Keep it short, engaging, and fun!

Here are some ideas for this block:

- **Rhymes and Poems:** Use math-themed songs or poems to introduce numbers, patterns, or simple operations.
- **Puzzles and Riddles:** Pose quick, simple math puzzles that get children thinking without the pressure of formal problem-solving.
- **Storytelling:** Weave math concepts into a captivating story. For example, a story about sharing cookies can introduce division.
- **Situation-Based Activities:** Ask questions like, "If we have 3 red pencils and 2 blue pencils, how many pencils do we have in total?" based on everyday scenarios.



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The goal here is to spark curiosity and activate their prior knowledge in a lively way

Block 2: Skills Teaching – Building Conceptual Understanding

This is where the magic of **conceptual understanding** happens! In this block, you'll introduce new mathematical skills through well-designed activities and structured instruction. It's crucial to connect the "how" (procedure) with the "why" (concept).

To truly ensure the understanding, one shall take note of:

- **Using Concrete Materials:** Bring out the manipulatives! Counting blocks, abacuses, play money, or even natural objects like pebbles can make abstract concepts tangible.
- **Incorporating Aids and Visuals:** Charts, diagrams, flashcards, and even drawing on the board can help children visualise mathematical ideas.
- **Demonstrating Clearly:** Model the new skill step-by-step, explaining your thinking aloud.

By using these tools, you help children grasp the underlying ideas rather than moving to mere rote memorisation.

Block 3: Skills Practice – Reinforcing and Fluency Building

Once a concept and a skill have been introduced, **skills practice** is essential for reinforcing understanding and building fluency. This is where children get to apply what they've learned in various situations.

Here's how you can facilitate effective practice:

- **Worksheets:** Design clear and focused worksheets that target the specific skill. Vary the types of problems to encourage different levels of thinking (procedural, conceptual, problem-solving, and reasoning).
- **Group Activities:** Encourage collaborative learning through group tasks where children can discuss, explain, and solve problems together.
- **Independent Activities:** Provide opportunities for individual practice, allowing children to work at their pace and build confidence.

The aim is to solidify their understanding and ensure they can confidently use the new skills.



Block 4: Math Game—Fun with Problem-Solving

End your math session on a high note with a **math game!** This block is all about reinforcing taught skills through fun, engaging activities. It’s a fantastic way to encourage problem-solving and ensure the learned concepts stick.

Consider these types of games:

- **Board Games:** Create simple board games that require counting, adding, or identifying patterns to move forward.
- **Card Games:** Adapt traditional card games to practice number recognition, sequencing, or basic arithmetic.
- **Interactive Digital Games:** Utilize age-appropriate educational apps or online games that reinforce mathematical skills.
- **Physical Games:** Games like “Math Hopscotch,” where children jump to the correct answer, can be highly engaging.

Table 4.1 is showing objectives, suggested strategies and schedule for each block in the four-block approach.

Table 4.1 Suggested Daily Blocks for Mathematics Teaching–Learning with Objectives, Strategies, and Time Allocation (National Curriculum Framework For Foundational Stage 2022, Pg. 122)

Blocks		Objectives	Suggested Strategies and Approaches	Suggested time
Blocks 1 & 2	Math Oral Talk	As warm-up activities for encouraging oral math	Open-ended/large group discussion Singing, poems, talk about children’s real life math experience, concept, oral calculation, reasoning	5–10 mins
	Skills teaching (Combining all strands)	Helping children to achieve mathematical skills through structured instruction/ activities	GRR/ELPS/ Problem-solving approach Conducting activities to build concepts, processes, application, strategies and reasoning	20–25 mins

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Blocks		Objectives	Suggested Strategies and Approaches	Suggested time
Block 3 & 4	Skills practices	Helping children to master skills through skills practice	Providing math tasks through workbooks or worksheets Individual, peer, group practice	15 mins
	Math game for reinforcing learning	Reinforcing taught skills through games Focusing on problem solving	Playing math games with children to reinforce learning and supporting children who are struggling	15 mins

By consistently implementing these four blocks, teachers can create a dynamic and effective mathematics learning environment that strengthens foundational numeracy skills in primary students.



ACTIVITY 4.3

Identify a topic from class 3rd NCERT Mathematics textbook and plan it using the **Four-Block Approach**.



CHECK YOUR PROGRESS 4.1

- How does the mode of representation facilitate thinking among learners?
- In what aspects do the three approaches discussed above connect?
- Fill in the blanks:
 - Oral math talk is a _____ activity.
 - The third block of the four-block approach deals with _____ the attained skills or concepts in various situations.
 - In experiential learning, reflection should be taken after _____.
- Choose the correct option.
 - Which of the following is not an aspect of the 4-block approach?
 - Oral math
 - Skill teaching
 - Writing skill
 - Skill practice

- b. Oral Math Talk should not involve:
 - i. Storytelling ii. Explanation
 - iii. Warm-up iv. Puzzles
- c. In the 4-block approach, which block emphasizes conceptual understanding?
 - i. Oral math talk ii. Skill teaching
 - iii. Skill practice iv. Math games
- d. A teacher blindfolded the student and gave her one object, then she asked her to describe its side's corners and faces while introducing the concept of 3D shapes. How does hands-on practice directly contribute to her mastering the concept?
 - i. It replaces the need for any theoretical understanding.
 - ii. It is primarily for memorizing steps rather than deep understanding.
 - iii. It helps in developing spatial understanding while exploring with objects and identifying features of objects and expressing them using language.
 - iv. It provides just a verification of concept by relating with the real world.

Answers:

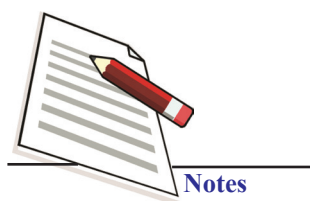
3. Fill in the blanks:
 - i. Engaging ii. Practicing iii. Concrete Experience
4.
 - a. (iii) Writing Skill
 - b. (ii) Explanation
 - c. (ii) Skill Teaching
 - d. (iii) It helps in developing spatial understanding while exploring with objects and identifying features of objects and expressing them using language.

4.4 ENHANCING PRIMARY MATHEMATICS TEACHING WITH MANIPULATIVES AND RESOURCES

Manipulatives and other resources are invaluable tools for teachers, helping to bridge the gap between abstract mathematical concepts and concrete understanding. They can be physical objects or digital tools, and their effective use can significantly enhance learning outcomes and foster a deeper engagement with mathematics.



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Manipulatives play a crucial role in reducing the abstractness of mathematical concepts, making learning more meaningful and accessible for students. By allowing students to interact directly with objects, they can break down complex ideas into manageable pieces and build a solid understanding. Resources can be categorised as material (e.g., charts, models, geoboards, measuring instruments), non-material (e.g., daily life activities, classroom discussions, language), and even human (e.g., teachers, peers).

As highlighted by the **NCF 2023**, “Resources enable students to comprehend concepts more effectively, as they connect verbal instruction with real experience, concretize abstract concepts, and develop curiosity and interest in learning.” This hands-on exploration facilitates mathematical understanding and thinking, helping students build connections between prior knowledge and new concepts. Much research shows that manipulatives, when used consistently and with guided instructions, improve conceptual thinking, enhance engagement, and are particularly beneficial for children with disabilities. Teachers themselves acknowledge that manipulatives simplify concept presentation and foster critical thinking and problem-solving skills.

4.4.1 Role of Manipulatives in Mathematics Teaching-Learning

Manipulatives are the resources used by the teacher or students to facilitate learning. They can be physical as well as digital. Manipulatives and other resources help teachers in teaching and learners in learning. In mathematics these manipulatives and resources play a significant role in reducing the abstraction of the concept and making learning meaningful. Resources and manipulatives used by students can be broken into pieces and built into a unit or concept. There is an exhaustive list of manipulatives, which can be categorised as material, non-material, and human. Objects like charts, models, blackboard geoboards, scales, measuring instruments, paper, and other concrete material can be termed as material manipulatives. A nonmaterial resource refers to an idea that one creates from exploring daily life activity, a teacher, a friend, discussion in the classroom, or language used in the classroom.

According to NCF 2023, “Resources enable students to comprehend concepts more effectively, as they connect verbal instruction with real experience, concretize abstract concepts, and develop curiosity and interest in learning.” Of course such tools help learners understand abstract concepts of mathematics by exploring the concept through hands-on activity, which facilitates mathematical understanding and thinking. They also help in building links with previous knowledge and new concepts. Moyer (2001) and Sowell (1989) stated that manipulatives improve conceptual thinking if used consistently with guided

instructions. Research comprehensively analyzed that use of resources in the classroom enhances engagement and learning, especially when there is an explicit connection between the instructions, manipulatives, and mathematical ideas, which are abstract in nature (Carbonneau, Marley & Selig (2013); Camelia Tjardra (2023)). Thus, it is advocated that manipulatives help in expansion and improve engagement of children with disabilities.

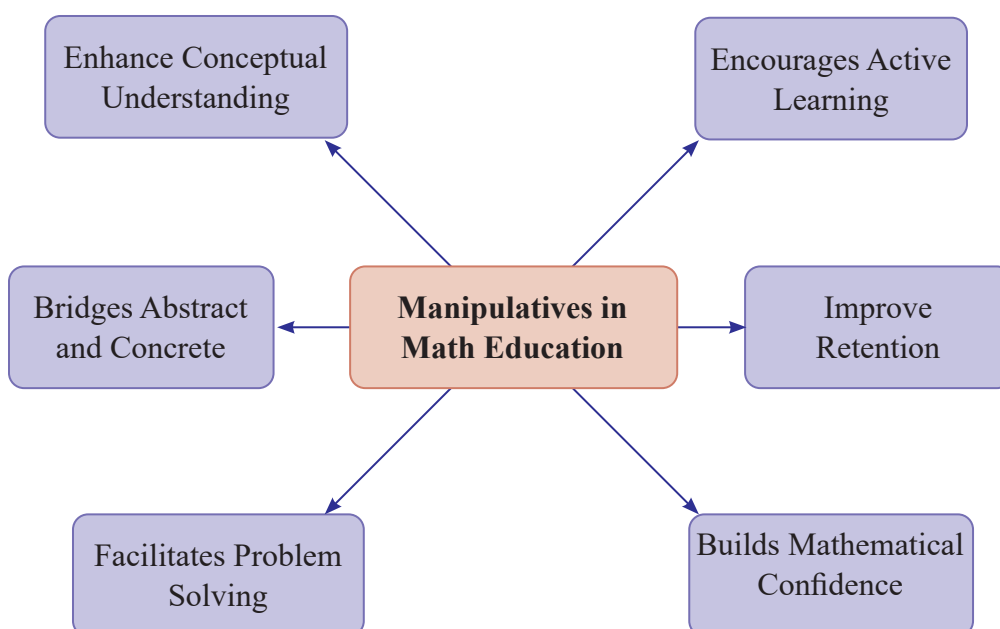


Fig 4.5 Role of Manipulatives in Mathematics Teaching - Learning

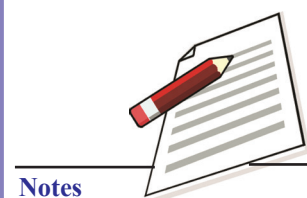
Teachers also acknowledged the use of manipulatives, stating that the use of resources makes presenting concepts easy as well as develops critical thinking and problem-solving.

Many of us use manipulatives to explain and explore various concepts. Their success depends on thoughtful selection and integration with instructions (Arthur J. Broody).

One should provide learners the opportunity to explore manipulatives freely. This free exploration helps learners understand the tool for learning and foster a deeper connection to the mathematical ideas they represent.

Why Manipulatives

1. **Concrete Experience:** Manipulatives provide a crucial opportunity for **concrete experience**. Learners engage in hands-on activities, which facilitates the development of various cognitive faculties and helps them



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build connections between different aspects of a concept. These transform learning experiences into something meaningful and comprehensive.

2. **Engagement and Motivation:** Engaging with diverse resources not only heightens a child's interest but also clarifies and simplifies the learning journey. In mathematics, manipulatives provide a tangible way to present abstract concepts, leading to increased **motivation and interest** in the subject.
3. **Inclusive Learning Through Resources:** The use of various resources and manipulatives allows teachers to cater to **diverse learning styles** within a classroom. They provide a **multi-sensory approach**, offering a platform to address the varied needs of all students in the class, ensuring that every learner has a pathway to understanding.

While manipulatives are powerful, their effectiveness isn't inherent. But at times the resource used doesn't sound effective or is misinterpreted by the learner. Why does it happen? Yes, such situations usually occur if manipulatives or resources are not selected wisely. At times, preparation with manipulatives lacked proper planning or was followed blindly. One should choose manipulatives purposefully, reflecting on the objectives of the lesson, relating to the concept, and also planning instructions while performing the activity. Manipulatives aren't inherently effective. Simply using manipulatives isn't enough; their true value comes from deliberate selection and skilled integration into the learning process.

Let's discuss two mathematics classroom situations.

Nalini, a grade 4 teacher, while teaching solid shapes, asked students to bring boxes of various shapes. She also had a good collection of various solid shapes. The next day she made groups of students and distributed the shapes to each group with the one they brought. Now she asked them to create something using these shapes. Later she made each group describe how and why they created this shape.

On the other hand, Sunita, a colleague of Nalini, started the chapter by defining all solid shapes and drawing the shapes in the notebook. Then, showing objects related to each shape, she demonstrated the faces and edges of the shapes.

According to you, which of the two teachers used manipulatives effectively? Why?

Considering the approaches of Nalini and Sunita, **Nalini** clearly used manipulatives more effectively.



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Why Nalini's approach was more effective:

- **Hands-on Exploration and Construction:** Nalini encouraged students to bring and use their own boxes of various shapes and then tasked them with creating something new. This allowed for **free exploration** and **active construction of knowledge**. Students weren't just passively observing; they were actively manipulating shapes, identifying their properties, and using them creatively. This aligns with the principle that learners should explore manipulatives freely to understand the tools and foster a deeper connection to the mathematical ideas they represent.
- **Conceptual Understanding through Discovery:** By asking students to describe *how* and *why* they created their shapes, Nalini prompted deeper thinking about the properties of solid shapes and their relationships. This approach moves beyond rote memorisation of definitions and encourages **conceptual understanding** through direct experience.
- **Engagement and Ownership:** Students were engaged because they were actively involved in the learning process, using materials they brought themselves, and working collaboratively in groups. This fosters a sense of **ownership** over their learning.

Why Sunita's approach was less effective in using manipulatives:

- **Passive Learning:** Sunita started by defining and drawing shapes, which is a more traditional, abstract approach. While showing objects later, her method was primarily a demonstration, with students as passive observers rather than active participants.
- **Lack of Exploration:** Students were not given the opportunity to explore the shapes themselves or discover their properties through hands-on activity. The focus was on identifying pre-defined features (faces and edges) rather than on deeper conceptual understanding through manipulation.
- **Limited Engagement:** This approach is less likely to engage diverse learning styles or cater to students who benefit from multi-sensory experiences.

The strategic integration of manipulatives and diverse resources is paramount for effective primary mathematics education. These tools are not mere accessories but essential elements that transform abstract mathematical ideas into tangible, comprehensible experiences. By fostering concrete understanding, boosting student engagement and motivation, and enabling

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inclusive learning through multi-sensory approaches, manipulatives empower teachers to cater to a wide range of learning styles and ensure that every child can build a strong foundation in mathematics. Ultimately, the thoughtful selection and skillful integration of these resources, as powerfully exemplified by Nalini's approach, are key to cultivating curious, confident, and capable young mathematicians.

Think and Reflect

Anita, a grade 5 teacher, took students to visit the monument. Is this visit a type of manipulative? What topics of mathematics as a teacher will you relate to Qutub Minar?

4.5 MATHEMATICS TEXT-BOOK AT PRIMARY STAGE

Most of the Indian classrooms revolve around textbooks. What is to be done and how it is to be done is directly based on the textbook assigned to the class. It provides a guide map to the teacher. The content and questions prescribed in the book are discussed and answered in the classroom interaction. Thus, we can say textbooks play a vital role in Indian classrooms' teaching-learning process. It is an easily available and important resource for students as well as teachers.

Mathematics textbooks are also an essential tool of teaching at the primary level, being one of the main resources for teaching in many remote areas. It is connecting sources between various stakeholders and the education system prevalent in the society or school system. It is an important source linking the classroom teaching and parents or scaffolders of the learner and other stakeholders. Parents or other elders can connect easily to what is going on in the class and accordingly guide the child.

Textbook, to teachers provides a model for interpreting the curriculum and syllabus designed for a particular class, stating the topics, examples of concepts, related activities, and exercises. But in today's globalized world, a textbook should not be the only source of teaching, though it could be a path director. With change in society the learner, their social milieu, and their abilities are also transforming, and the teacher should be the one connecting unit that could bridge the gap between the prescribed information and the social milieu of the learner. She should present the concept given in the textbook in such a way that each learner finds the scope of learning and exploring.

However, before going to book, one should go through the curriculum's overall philosophy and the goals for which the textbook has been designed; otherwise,

it is very difficult to impress the objectives and the philosophy of the prescribed. With NCF 2023, significant transformations have been incorporated in the textbooks, new activities, new ideas, and different types of questions, especially divergence, which got space in the system.

Also in the classroom process, textbooks should not be the only source of teaching and learning. It can act as a guiding path. The role of teacher becomes very crucial while imparting the prescribed textbook knowledge, as she must go beyond the text, relating it to the local context of the learner. Rather than following the text blindly, one needs to blend the content in social context and daily life activities such that learning should become meaningful and relevant in context. As in the class 3 textbook, the chapter **Toy and Joy deals** with 3D shapes. Here students can be allowed to make something using objects of various shapes. They can be motivated to use soil or clay to make something. Even various types of boxes and bottles can be collected from students and can be used to explore shapes. Similarly, a project can be given to learners to explore the old building or monument in their nearby area and look for and list various shapes used to create it.

Similarly, lots of puzzles are part of textbooks. But as a teacher you can explore for more relevance in the context of your learners. The local artwork, heritage, and occupation can be part of learning and exploring in terms of pattern, measurement and shapes. While teaching measurement, one should describe how local terms are related to standard terms and how one can be converted into another. Issues like improper use of terms should be discussed in the class to avoid confusion. Like milk being liquid is measured in litres, but usually it is asked in kg.

Similarly, in class 4, in the chapter “Thousands around us,” the concept of langar (a community lunch) is taken, which is a common practice in temples and Gurdwaras. Learners can be asked to explore when such an occasion is taking place in their locale. This is how one can go beyond books and relate to daily life.

Textbooks have recently been changed. Select one common chapter from both books (old and new) and identify the changes made in the new textbook compared to the old one. Reflect on the need for changes being made.

Characteristic of a good book

1. Textbooks, especially at the primary level should be colorful, attractive, and easy to understand.

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2. Language used in the book should be easy to comprehend for the learner.
3. It should not carry too much content. It should be child-friendly. Content presentation should provide scope to think, talk and work mathematically.
4. Rather than explanation, it should also provide opportunities to think critically and facilitate discussions and observations.
5. A constructive textbook allows learners to express their thoughts and encourages them to engage with learning material as well as reflect on their learning.
6. It should suggest what resources and strategies a teacher can use with respect to the philosophy of the book.
7. It should carry a variety of activities and questions that foster mathematical thinking rather than merely mugging up the facts and algorithms.

Think and Reflect

According to you, how are the new NCERT books better than the previous ones?



ACTIVITY 4.4

Take two books of the same class and compare the two? List 5 to 7 points that are good in each book and 5 to 6 points that you feel need to be changed and why.



CHECK YOUR PROGRESS 4.2

1. What makes experiential learning a step ahead of learning by doing?
2. **Fill in the blanks:**
 - a. The textbook is a vital _____ of Indian classrooms, but it should not be followed blindly.
 - b. A good book should provide opportunity to observe, _____ and _____ to the learner.
 - c. Manipulatives and diverse resources act as essential elements that _____ abstract mathematical ideas into tangible, _____ experiences.

- d. Use of puzzles, brain teasers and riddles in teaching and learning mathematics provides a platform for _____ and _____ thinking.

3. **Choose the correct option:**

- a. In what way can textbooks promote critical thinking?
- (i) By strictly dictating interpretations of complex topics.
 - (ii) By requiring students to memorize large amounts of data without context.
 - (iii) By offering questions that encourage analysis, comparison, and evaluation of information.
 - (iv) By presenting only one viewpoint as absolute truth.
- b. What is a potential drawback of relying solely on textbooks for learning?
- (i) They are generally too interactive and distracting for focused study.
 - (ii) They may not always be up-to-date with the latest research or developments in rapidly evolving fields.
 - (iii) They do not offer a structured learning path.
 - (iv) They always provide the most current and up-to-the-minute information.
- c. What is the primary purpose of a textbook in education?
- (i) To serve as a comprehensive and structured source of information on a specific subject.
 - (ii) To offer a platform for social media interaction among students.
 - (iii) To replace teachers and classroom instruction entirely.
 - (iv) To provide entertainment through engaging stories.

Answers:

2. Fill in the blanks:
- (a) Resource
 - (b) Think and reflect
 - (c) Transform, comprehensible
 - (d) Mathematical, critical



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3. (a) (iii) By offering questions that encourage analysis, comparison, and evaluation of information.
- (b) (ii) They may not always be up-to-date with the latest research or developments in rapidly evolving fields.
- (c) (i) To serve as a comprehensive and structured source of information on a specific subject.



4.6 LET US SUM UP

- Classroom teaching should weave around children's knowledge construction, for which they should be provided with hands-on experiences.
- Experiences, if facilitated by reflection and various modes of representation, namely picture then symbol, help children articulate new ideas and add to their interest.
- Manipulatives and resources make the classroom lively and facilitate learning. Selection of resources and manipulatives should be done purposefully.
- Mathematics textbooks at the primary level are an essential resource in the classroom but not the only source of the teaching-learning process.



4.7 END EXERCISES

1. Comment on the three approaches discussed above.
2. Why does giving experience facilitate learning?
3. Compare NCERT's new books with any private publisher's book and reflect on the philosophy behind both of them.
4. Discuss the role and importance of manipulatives in teaching shapes in primary classes.

4.8 REFERENCES

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